Towed-Sensor Surveillance UAV

AMERICAN UNIVERSITY OF THE MIDDLE EAST

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1.0 EXECUTIVE SUMMARY

Objective. Machine Masters will design, build and fly a hand-flown electric aircraft that carries, tows and protects the 2027 sensor payload, with one objective: the highest total score at Tucson in April 2027.

Approach. The rules became 19 subsystem requirements; 162 aircraft were scored on 120 draws of every uncertain input against 5,000 rival fields calibrated on 2022–26 fly-offs. Two findings set the design. Completion pays 4.5 of the 7.5 mission points, so reliability carries 55 % of the trade weight. The declared sensor mass sets the Ground Mission, every Mission 2 unit and the Mission 3 ballast cap; declaring it at technical inspection is worth +0.42 points.

Design. A 1.82 m span, 1.00 m² high wing with Fowler flaps (trimmed $C_{L,\max}$ 1.93), conventional tail, fixed tricycle gear and one 8S tractor lands its 14.6 kg Mission 2 at 11.9 m/s. The plan waives Mission 1, flies one 7.4 kg unit five laps in 231 s, tows 3.4 kg for 6 laps and drops the declared sensor from 1.524 m: a mean of 6.76 of 7.5 and a 0.313 chance of the field's highest mission total, against 0.128 for our first-iteration design. A slotted 1.2 m² wing is the dated fallback.

Execution. Six sub-teams build two airframes from laser-cut, printed and carbon-tube parts, retire the top risks by 15 January and freeze the design on 19 February.

DESIGN AT A GLANCE / VARIANT A

WINGSPAN	WING AREA
1.82 m	**1.00** m²
M2 GROSS MASS	STALL, M2
14.6 kg	**11.9** m/s
DECLARED SENSOR	M2 FIVE LAPS
3–8 kg	**231** s
M3 LAPS AT	GM DROP
6 × 3.4 kg	**1.524** m
MISSION SCORE	P(TOP OF FIELD)
6.76 / 7.50	**0.313**

2.0 MANAGEMENT SUMMARY

2.1 Team Organization

Machine Masters has 21 members in six sub-teams, organized by function to mirror how the aircraft is scored (Fig. 02). The team lead owns schedule, budget and rules compliance; the chief engineer owns the configuration and the requirement list, and only the chief engineer may change a requirement. Each sub-team owns one scored subsystem and its tests (Table 01); Sensor Systems exists because the sensor, drum and container touch three of the four mission scores. Every decision is a card stating its evidence, the test that would overturn it and its freeze date, signed at PDR, the A/B gate and configuration freeze.

2.2 Schedule and Milestones

The schedule (Fig. 01) is built backward from the fly-off and gated by decisions. Parts common to both variants are ordered first, so the A/B decision costs one wing, never a second program. The prototype flies in early January, before the A/B gate, so the single airframe presented at the contest is never the first one flown; its weigh-in and lap times recalibrate the model before the design report.

TABLE 01 Sub-teams, what each owns and the skills each requires. Every scored subsystem has exactly one owner.

SUB-TEAM	OWNS	REQUIRED SKILLS
Aero & Performance	Sizing, lap model, stability, CFD	Python modelling; XFoil, AVL, OpenFOAM
Structures & Manufacturing	Wing, Fowler tracks, fuselage, tests	CAD; laser cutting, printing, foam cutting
Propulsion & Avionics	Motor, propeller, pack, fuse, radio	Thrust-stand testing; LiPo safety; ExpressLRS
Sensor Systems	Dart, lights, drum, container, test box	Embedded C; servo drives; cushioning design
Flight Operations	Pilot, observer, crew, checklists	AMA-rated piloting; timed staging drills
Business & Logistics	Budget, purchasing, travel, documents	Lead-time planning; cost control; writing

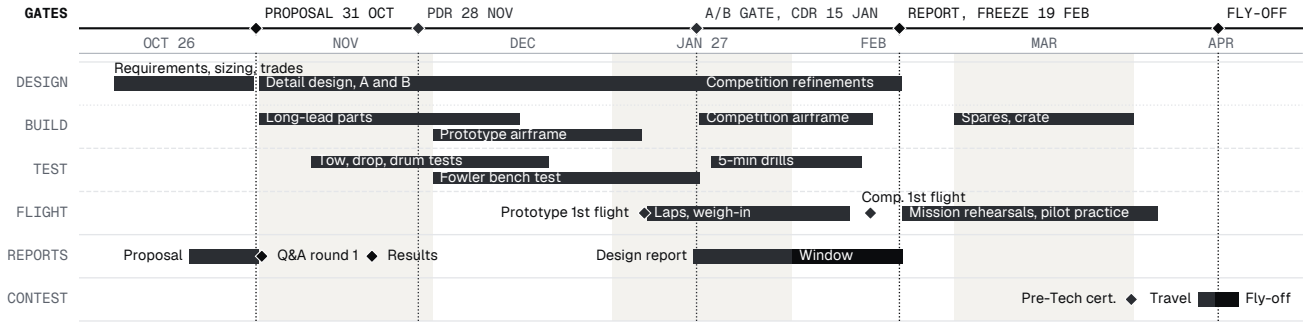


FIG. 01 The schedule is gated by decisions: long-lead parts first, the Fowler bench test decides Variant A or B at the 15 Jan gate, the prototype flies in early January, and the configuration freezes with the design report on 19 Feb. Ink: dates fixed by the rules; graphite: team plan.

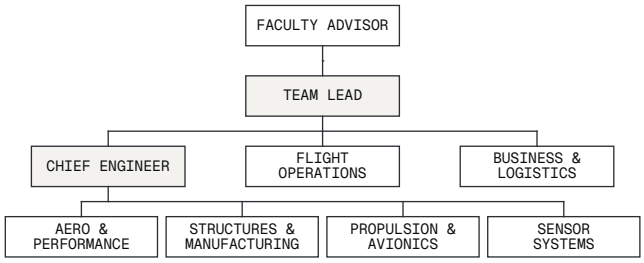


FIG. 02 One owner for every scored subsystem; Sensor Systems is a peer of the airframe teams.

TABLE 02 Program budget for the whole year, every expected cost including travel (USD). List: itemized purchase list at vendor prices; quote: retrieved 31 Oct 2026; avg.: online average; est.: estimate.

ITEM	BASIS	COST
Airframes, B wing, Fowler test, rigs	list, est.	3,260
Propulsion, radio, servos, 8S packs	list	4,800
Sensor, drum, containers, tungsten	list, avg.	1,320
Spares, consumables, crate	list	610
Airfare, Kuwait–Tucson, 7 people	quote	10,820
Lodging, van, crate baggage	quote, est.	4,050
Fly-off fee, AIAA dues	rules	1,330
Contingency, 15 %		3,930
Total		**30,120**
Funding, university	confirmed	32,170

2.3 Budget

Table 02 covers the year: the itemized purchase list, a second airframe and backup wing, 8 kg of tungsten at the $66/kg online average, and travel for 7 at $1,550 per quoted fare.[1] A 15 % contingency covers a crash rebuild; the confirmed university allocation of 10,000 KWD ($32,170) covers the total.

3.0 CONCEPTUAL DESIGN APPROACH

3.1 Mission Requirements Decomposition

Each mission score and rule became a numbered subsystem requirement (SR) that the trades and tests cite (Table 03); † marks requirements the team added beyond the rules.

3.2 Scoring Model and Sensitivity Study

The mission score sums four terms, each a completion credit plus a ratio to the best result in the field, our own included:[2]

$$S_{\mathrm{M}} = M_1 + \left[1 + \frac{(W/t)}{(W/t)_{\max}}\right] + \left[2 + \frac{NW_{\mathrm{s}}}{(NW_{\mathrm{s}})_{\max}}\right] + \left[0.5 + \frac{W_{\mathrm{s}}h}{(W_{\mathrm{s}}h)_{\max}}\right] \le 7.5 \tag{1}$$

Completion is worth 4.5 points and the bonuses at most 3.0, yet in 2026 only 17 of the 83 teams that

[1] Lastminute.com quotes, KWI–TUS, 12–20 Apr. 2027, retrieved 31 Oct. 2026.

[2] AIAA, *2026–27 Design, Build, Fly Rules*, Draft, 2 Sep. 2026, Secs. 3.1–3.3, 6.3.

TABLE 03 Mission requirements decomposed into subsystem requirements, cited by the trades (Table 04) and tests (Table 09); † team requirement beyond the rules.

MISSION	SCORE	KEY CONSTRAINTS	DERIVED SUBSYSTEM REQUIREMENTS
M1 Staging	$M_1 = 1.0$	3 laps in 5 min; may be waived and credited by a successful M2	SR-01 static margin 16 % MAC in every loading†; SR-02 all checks inside the 5-min staging
M2 Delivery	$M_2 = 1 + \frac{W/t}{(W/t)_{\max}}$	Sensor in container plus simulators, each at the declared maximum; TOP up; 5 laps	SR-03 bay for a 288 × 303 × 303 mm container; SR-04 positive restraint; SR-05 CG inside marked limits in every loading
M3 Sensor	$M_3 = 2 + \frac{NW_s}{(NW_s)_{\max}}$	Deploy before the first downwind turn; OFF, SOLID, FLASHING, OFF; stow before landing; sensor ≥ 152.4 mm (6 in)	SR-06 internal permanent drum, tow ≥ 2.73 m; SR-07 three lights, modes over the line, visible in sun; SR-08 stable tow; SR-09 own COTS battery; SR-10 line clear of propeller, tail†
Ground	$GM = 0.5 + \frac{W_s h}{(W_s h)_{\max}}$	6 drops on concrete, $h \le 1.524$ m (60 in); ring to bottom < 381 mm (15 in); 5-min rigging; no damage	SR-11 container absorbs $W_s h$, sensor rated to 150 g; SR-12 tool-less O/E rigging, flush fittings; SR-13 test box drives all three modes
System	–	Span ≤ 1.829 m (6 ft); TOGW < 24.95 kg (55 lb); ≤ 100 Wh; no autopilot, GPS or camera	SR-14 pilot-in-the-loop only; SR-15 positive latches, no magnets; SR-16 energy for M2 at full payload; SR-17 §4.2.5 fail-safe, range check; SR-18 structure for n = 3.24 and tip lift; SR-19 ESC, pack cooling†

passed technical inspection completed every mission.[3] The declared sensor mass W_s appears three times: it is the Ground Mission product, it sets every Mission 2 unit, and it caps the Mission 3 ballast. Winning mission totals ran 5.61–6.80 from 2022 to 2026.

Each design is therefore built and flown 120 times with every uncertain input drawn (mass, $C_{L,\max}$, drag, efficiency, density and a lap factor from 24 measured flights), declares its sensor at inspection, and meets 5,000 simulated fields. Fig. 03 is the sensitivity study about the selected aircraft. Lap time dominates: laps 10 % slower cost 0.54 points. Density, efficiency and airframe mass follow (0.22, 0.22 and 0.16 per 10 %). Both races are energy-bound (77.9 of 78.1 Wh used in Mission 2). Wing area is at its optimum: ±10 % moves the mean by under 0.03. Fig. 04 shows the largest lever: a sensor mass frozen at design peaks at 6.33 (4 kg), because a heavy frozen unit fails Mission 2 on heavy airframes; declared at inspection, the same aircraft averages 6.76.

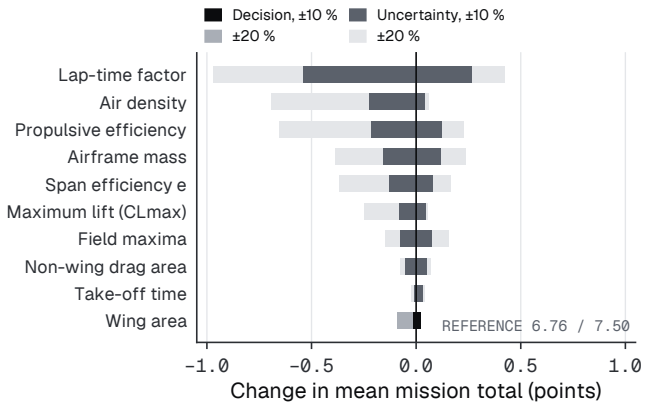


FIG. 03 Lap time, density, efficiency and airframe mass move the score most; wing area is at its optimum. Mean of 120 paired draws, Variant A; usable energy equals efficiency.

3.3 Configuration Trade Studies

Trade weights come from the sensitivity study: completion 55 %, lap performance 17 %, high lift 10 %, airframe mass 7 %. Table 04 gives each decision its number. The team's starting reference, the long-wing RQ-4 family, was run in OpenFOAM:[4] at 1.82 m span its 0.23 m² wing stalls near C_L 0.9, a 25 m/s landing at 7 kg, so its proportions were dropped. Fig. 05 screens 162 combinations of area, flap, declared range and

[3] AIAA, "DBF Competition Summaries," 2022–2026 (513 team results).

[4] OpenFOAM 14, RANS k–ω SST; indicative, used for maximum lift only.

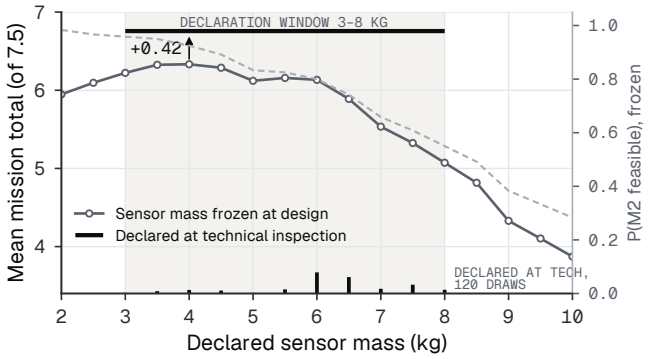


FIG. 04 Declaring at technical inspection is worth +0.42 over the best frozen mass; frozen at 8 kg, Mission 2 is feasible on only 0.55 of airframes. Bars: masses the 120 draws declare.

TABLE 04 Configuration decisions and their evidence, in mean mission points (propulsion row: P(top)).

DECISION	SELECTED	EVIDENCE
Planform	**Straight**	Delta −1.69; biplane −0.32
Flap	**Fowler**	1.00 m²; $C_{L,\max}$ 1.93
Airfoil	**SD7037**	Best of 13; trip-tolerant
Wing	**High**	Tow line and bay clear
Tail	**Conventional**	Rule fail-safe throws
Gear	**Fixed trike**	Retract +0.01 at best
Power	**One 8S**	12S −0.02; twin −0.05
Tips	**Plain**	Winglets −0.01 to −0.05
Sensor	**At tech**	+0.42 (Fig. 04)

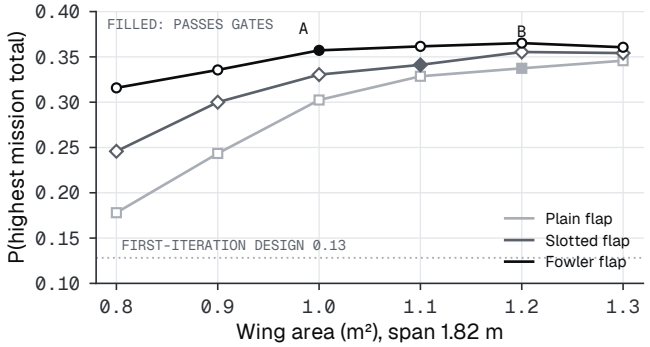


FIG. 05 Area substitutes for flap complexity: the Fowler flap reaches the leading probability with the smallest wing. 8S designs, best declared range per point.

propulsion: the high-lift flap buys a smaller wing, and the 1.0 m² Fowler wing leads the designs that pass the reliability gates.[5] The drum trade (Table 05) is weighted by points at stake.

[5] Flap data: R. C. Platt, NACA Rept. 534, 1935; C. J. Wenzinger and T. A. Harris, NACA Rept. 664, 1939; method validated on NACA TN 663.

TABLE 05 Deployment-and-recovery trade (scores 1–5). Boom, wireless link, bridle, pulley and spring reel failed the rules or the 20 s deploy window.

CRITERION	WT.	CLOSED DRUM	OPEN DRUM	FREE SPOOL
Recovery reliability	0.75	5.0	4.4	3.6
Manufacturability	0.15	3.1	5.0	1.7
Weight, volume	0.04	5.0	4.9	2.4
Signal path	0.03	5.0	5.0	5.0
Deploy, stow time	0.03	4.5	5.0	5.0
Weighted total	**1.00**	**4.70**	**4.55**	**3.39**

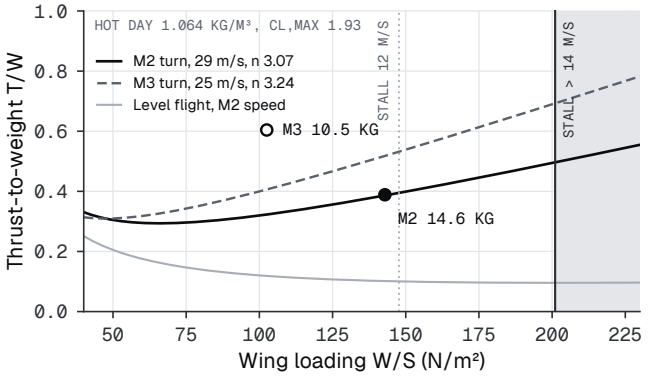


FIG. 06 Mission 2 is sized by the race turn at full thrust; the Fowler flap keeps its 14.6 kg landing under 12 m/s.

3.4 Preliminary Design and Sizing

3.4.1 Sizing method Sizing uses the hot-day density of 1.064 kg/m³[6] and a lap model of the 152.4 m (500 ft) legs, 180° turns and downwind 360°. Speed and bank are optimized under stall (≤ 14 m/s), turn lift, load factor (≤ 4), battery power and the 100 Wh cap, with the measured ×1.15 lap factor. In Fig. 06 the Mission 2 point sits on its 29 m/s, n = 3.07 turn line at full thrust, inside the 12 m/s stall line; Mission 3 keeps thrust margin and is limited by energy.

3.4.2 Aircraft concept Variant A (Fig. 07, Table 06) is a straight wing of aspect ratio 3.31 over a 0.345 × 0.355 m box fuselage, with a conventional tail on a 1.05 m arm. Bay and tow exit sit on the centre of gravity, so every loading flies at 16 % static margin (AVL: $C_{m\alpha}$ -1.07, $C_{n\beta}$ 0.082, $C_{l\beta}$ -0.012 per rad).[7] Variant

[6] 10th percentile at the daily high, 2,878 hourly KTUS observations, 15–18 Apr. 1996–2025, reduced to TIMPA (667 m). Iowa Environmental Mesonet, "ASOS Data Download," retrieved 11 Oct. 2026.

[7] M. Drela and H. Youngren, AVL 3.52; M. Drela, XFoil 6.99; MIT.



FIG. 07 Variant A: a 1.00 m² Fowler-flap high wing over a box fuselage sized around the container; the payload bay (dashed) and the tow exit sit on the centre of gravity, so every loading flies the same.[team GA drawing]

TABLE 06 Preliminary design parameters, Variant A. Gross masses are medians of 120 draws; empty mass is estimated from 14 top-3 aircraft.

PARAMETER	VALUE	UNIT
Span / area / aspect ratio	1.82 / 1.00 / 3.31	m, m²
Mean chord / tail arm	0.555 / 1.05	m
Empty mass, with pack	6.96	kg
Gross mass, M2 / M3	14.6 / 10.5	kg
Trimmed landing $C_{L,\max}$	1.93	–
Stall, M2 / M3	11.9 / 10.1	m/s
Static margin	16	% MAC
Thrust, AT7215 / 20×15E	101	N
8S 3300 mAh pack	97.7	Wh
Fuse, 58 V rated	80	A
Tow line, minimum	2.73	m

B changes only the wing (1.2 m², slotted), tail and boom.

3.4.3 **Sensor and D&R system** The sensor is a tungsten-ballasted dart, about 200 × 70 mm at 6 kg, rated to 150 g and towed from an eye above its centre of gravity so it flies level.[8] Its own labelled 1S pack drives three LED arrays of about 220 cd. A servo-driven closed-loop drum with a slip ring carries the line and, over it, the mode signal from the receiver in the airplane; one button steps OFF, SOLID, FLASHING. Stowing takes 16.5 s at 3.4 kg and 24.6 s at 8 kg, so Mission 3 flies lighter. The aircraft is flown pilot-in-the-loop at all times: no autopilot or flight controller is carried, and stabilization is limited to optional rate gyros.

[8] J. Barrowman, "Theoretical Prediction of the Center of Pressure," NARAM-8, 1966.

TABLE 07 Mission targets and predicted scores, Variant A: means of 120 draws against 5,000 simulated fields; ranges P10–P90.

MISSION	TARGET	SCORE
M1	Waived; credited by the first M2	1.00
M2	One 7.4 kg unit; 5 laps, 231 s (158–287)	1.46
M3	6 laps at 3.4 kg (5–9 laps)	2.85
GM	6.25 kg declared from 1.524 m (60 in)	1.44
Total, mean (P10)		**6.76 (6.25)**
P(highest mission total)		**0.313**

3.4.4 **Container and rigging** The container is built once for the top of the declared range: 41 mm of polyethylene foam keeps a 150 g sensor intact from 1.524 m with the ring under 381 mm from every face.[9] Pads are re-cut at inspection for the declared mass, simulators carry adjustable ballast within 28 g, and rigging uses flush, tool-less O/E fittings. All hatches and the container lid use positive mechanical latches; magnets are never the latching method.

3.5 Mission Design Approach and Scoring Targets

Every plan protects completion first (Table 07). **M1:** waived; a successful Mission 2 credits it, and the saved attempt lifts the chance of finishing Mission 3 from 0.40 to 0.72. **M2:** one unit at the declared mass, 14.6 kg gross, at 29 m/s with 71° turns (38.9 s laps); simulators only if the weigh-in leaves margin. **M3:** deploy with lights OFF on the upwind leg, SOLID after

[9] U.S. Dept. of Defense, *MIL-HDBK-304B, Package Cushioning Design*, 1978.

lap 1, FLASHING after lap 2, then OFF and stow before landing; ballast is chosen so the stow fits the window: 3.4 kg, 6 laps of 39.9 s. **GM:** 6.25 kg from 1.524 m gives 9.53 kg m against a simulated field best of 11.4 kg m; rigging drilled under 240 s. **Switch rule:** if the Fowler section gives trimmed $C_{L,\max}$ below 1.75, or jams more than once in 100 cycles, by 15 January, Variant B is built; it trails A by 0.018 in P(top) when the flap works and leads by 0.044 when it does not.

4.0 MANUFACTURING PLAN

4.1 Preliminary Manufacturing Flow

Airframe, sensor and Ground Mission lanes run in parallel and meet at integration (Fig. 08); every part is bench-tested in its lane first, and the prototype, built from the same files, flies before the competition airframe. The team's own tools cover every process (FDM printers, laser cutter, CNC router, hot-wire foam cutter); moulded composites are avoided by using pultruded carbon tubes for spars and booms.

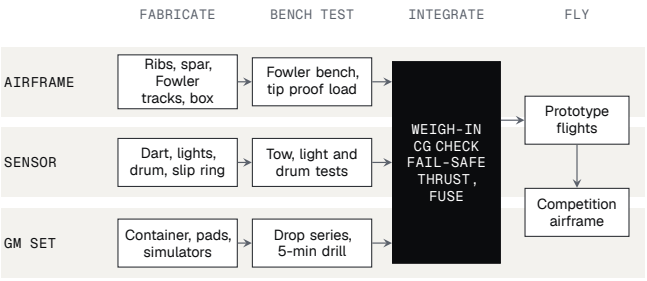


FIG. 08 Airframe, sensor and Ground Mission lanes are each proven on the bench before they meet at integration.

TABLE 08 Critical processes, why each matters to the score, and how it is made and proven.

PROCESS	WHY CRITICAL	METHOD AND PROOF
Fowler tracks	Sets $C_{L,\max}$	Printed carriages; bench
Wing	Tip lift, n 3.24	Ply ribs, carbon spar
Drum, slip ring	Line and signal	Servo drive; cycling
Tungsten dart	150 g, level tow	Potted; tow, drops
Container pads	Absorb $W_s h$	CNC-cut foam; drops
Power harness	80 A, 58 V fuse	10 AWG; duty run

4.2 Critical Processes and Technologies

The Fowler track is the one technology the team has never built, so it is made first as a test section, and its result chooses the variant (Table 08). The drum and dart are cycled and dropped on the bench long before they fly. The harness uses a 58 V-rated fuse because the common 100 A blade fuse is rated for only 32 V, below a charged 8S pack's 33.6 V.

5.0 TEST PLANNING

Tests are ordered by the risk they retire; each has a pass criterion from Table 03 and a dated slot (Table 09); Table 10 maps the top risks to them. Bench tests close before the A/B gate; flights then build to full-mission rehearsals. Every flight follows a written checklist, including the §4.2.5 fail-safe and range check, and logs power and lap time on a non-RF logger; each result recalibrates the model before the sensor mass is declared.

TABLE 09 Component, ground and flight test plan; failing a criterion triggers the alternative named in Section 3.0.

TEST	PASS CRITERION	DATE
Ground and component		
TC-01 tow, 3–8 kg	No sustained swing	Nov
TC-06 thrust, fuse	$\geq$ 95 N; fuse holds M2	Nov
TC-02, 03 drops, sun	Intact; read at 200 m	Dec
TC-05 drum cycles	98 % recovery	Dec
TC-07 Fowler section	$C_{L,\max} \geq 1.75$	15 Jan
TC-04 5-min drills	Load, attach $<$ 240 s	Jan
Tip lift, fail-safe	§4.2.4, §4.2.5 pass	Feb
Flight		
TC-08 weigh-in, CG	Fit $\pm$15 %; CG $\pm$3 %	Jan
TC-09 handling, laps	Model $\pm$15 %	Jan
Stepped M2 payload	7.4 kg, 5 laps $<$ 300 s	Feb
Full M3 sequence	Deploy, 3 modes, stow	Mar
Hot-day rehearsal	All missions, 5 % reserve	Apr

TABLE 10 Top risks: likelihood (L), consequence (C), retiring test.

RISK	L	C	RETIRED BY
Fowler underdelivers	M	H	TC-07; B wing
Dart swings in turns	M	H	TC-01; eye over CG
Stow costs a lap	H	M	TC-05; lighter dart
Fuse opens at take-off	L	H	TC-06; slow-blow